

**COMBINED SCIENCE
CHEMISTRY NOTES**

2019

DISCOVERING COMBINED SCIENCE BOOK 1-2



TOPICS

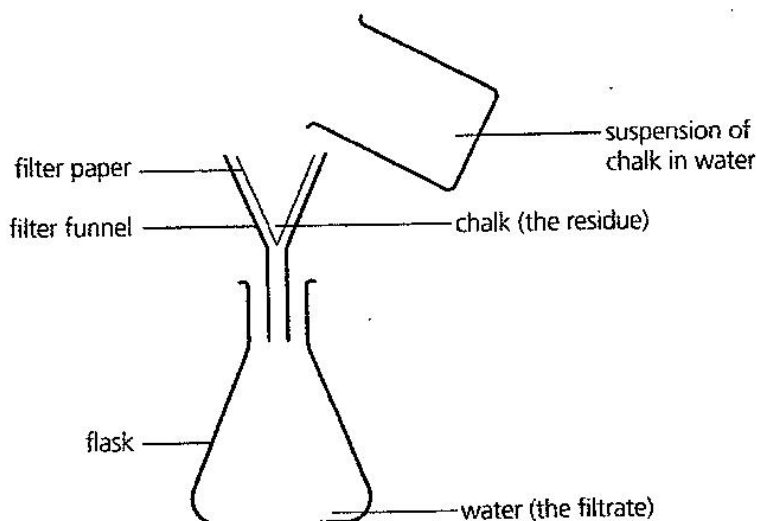
- SEPARATION
- MATTER
- ACIDS, BASES AND SALTS
- OXIDATION AND REDUCTION
- INDUSTRIAL PROCESSES
- ORGANIC CHEMISTRY

SEPARATION OF MIXTURES

METHODS OF SEPARATING MIXTURES

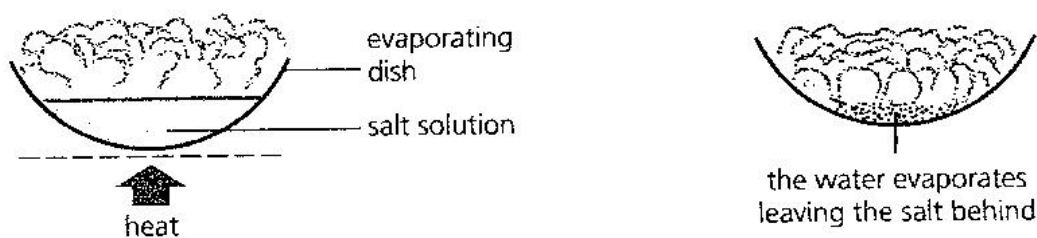
Filtration

- It is a process of passing a fluid through a filter to remove solid particles.



- Is used to separate insoluble solids from liquids e.g. a mixture of soil and water.
- A filter is placed in a funnel, on a flask and the mixture is poured on the filter paper. The liquid filtrate passes through the filter paper and is collected. Insoluble substances are left as residue on the filter paper after filtration.
- Filtration can be used in the following process
 - Water treatment i.e. filtration removes all suspended solids like plants and animal matter from water before it is made available for use.

Evaporation



- Is used to separate a solution in which the solid is dissolved in the liquid.
- The solution is heated so that the solvent evaporates and the solid is left behind after all the liquid has evaporated.
- Evaporation is used in the following processes
 - Formation of sugar crystals – evaporation plays an important role in the extraction of crystals of sugar from sugar cane juice. The clear juice is piped into evaporators which reduce the mixture to thick sugary syrup.
 - Formation of ammonium nitrate crystals from their solutions. Excess water in the ammonium nitrate solution is evaporated to produce a saturated solution. The ammonium nitrate crystallizes out of the solution.
 - Table salt processing – shallow ponds of sea water are left to dry in sunshine until all water has evaporated.
 - Salting peanuts – g

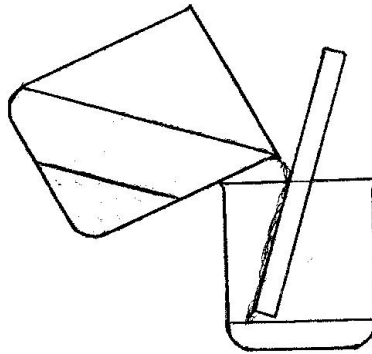
Winnowing



- Winnowing is used to separate solids with different densities. Lighter materials will blow away while the heavier material will fall to the ground.
- A winnowing basket is shaken vigorously from side to side. The less dense come to the top and are blown off while the heavier remain in the container
- Winnowing can be used in the following processes
 - Grain separation – winnowing by wind is a method of separating usable grain from husk (chaff). To remove unwanted components, the mixture is thrown into

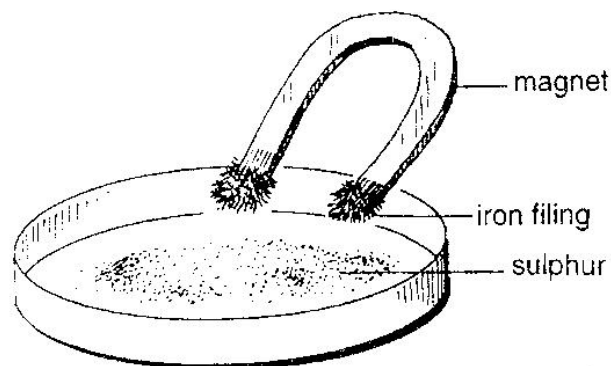
air where wind blows the husks away and the heavier grain fall back into the basket.

Decanting



- Decanting is a process used to separate immiscible liquids that have different densities. A distinct layer between the two constituencies is formed.
- The layer closer to the top of container – the less dense of two liquids is poured off, leaving the more dense liquid of the mixture behind.
- Decanting can be used to
 - Separating a mixture of oil and water – oil floats on top of water and is carefully poured off
 - Separating a mixture of whey from milk – whey floats on top of

Magnetism



- Magnetism is used to separate a mixture of solids in which one of the components has magnetic properties.

- Iron filings are attracted by the magnet, so they cling to the magnet, while the sulphur powder remains behind on the paper.
- Magnetism can be used in the following processes
 - Separation of metallic objects from grain before grinding – metal pieces can cause damage to the processing machinery. Magnets are used to attract any metallic objects before grinding.
 - Separation of metallic wastes for recycling – electromagnetic cranes are used to attract large loads of scrap iron and steel from wastes.

MATTER

STATES OF MATTER

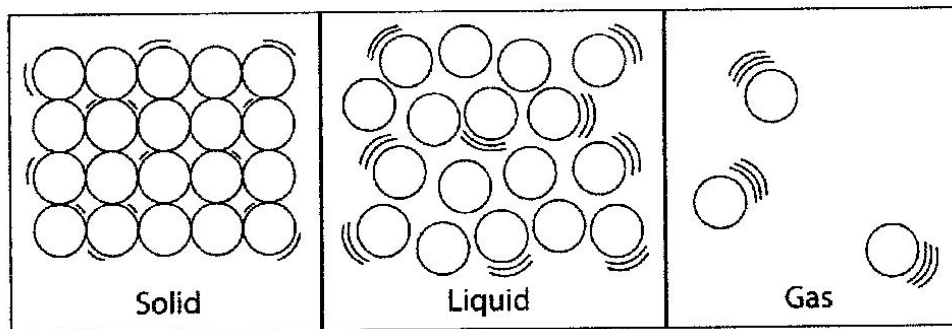
- Matter is anything that has mass and occupies space. It can be solid, liquid or gas.

Kinetic Theory

- The kinetic theory of matter states that matter is made of tiny particles that are always in constant motion.
- The behavior of these particles differs in the 3 phases

The three states of matter

- The properties of solids, liquids and gases can be explained by the kinetic theory.



Solid

- The particles in a solid are arranged in a fixed pattern or lattice. This accounts for the high density of solids.
- Strong intermolecular forces hold them together. This explains why solids have fixed volume and shape.
- The particles have very little kinetic energy and they only vibrate in fixed positions.
- Properties include definite shape, fixed volume, cannot flow, cannot be compressed and high density

Liquid

- The particles occur in clusters with the molecules slightly further apart compared to that of a solid. This accounts for the high density of liquids and the tendency of liquids to form droplets.
- The particles have some kinetic energy. The particles are free to move about between clusters but confined within the vessel containing it due to attractive forces between them. This explains why liquids have fixed volumes but take the shape of the vessels containing them.
- Properties include no fixed shape, less dense than solids, cannot be compressed, fixed volume and can flow.

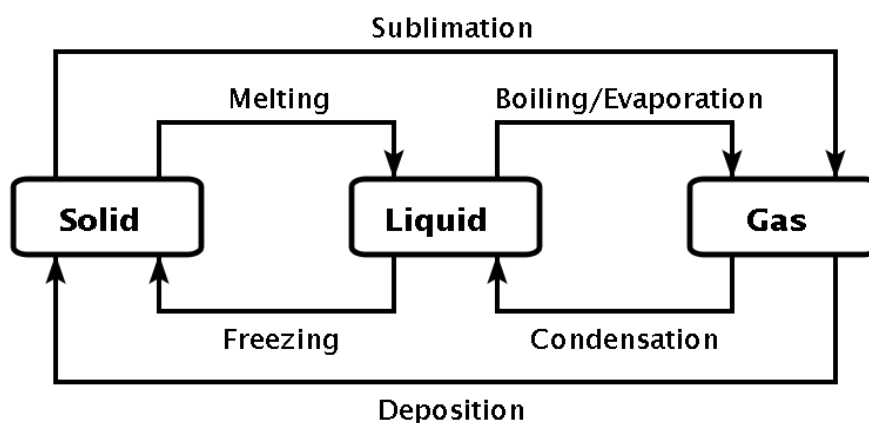
Gas

- The particles are very far apart in that the molecules will occupy any available space. This accounts for low density of gases.
- The particles have lots of kinetic energy. The particles move at high speed, independent motion in random manner. Negligible attractive forces exist between them. This explains why gases have neither fixed volume nor shape.

- Properties include no fixed volume, shape and size, can flow and highly compressible.

Change of states

- If temperature is increased or reduced, matter changes its state. This is because a temperature change will affect the amount of kinetic energy.
- If a solid is heated, the energy will make the particles vibrate faster until the forces that hold them together are weakened and the particles move apart.
- If a gas cools, the particles lose energy and come closer together. Eventually strong forces develop between them.
- All changes of state involve an increase in kinetic energy or a decrease in kinetic energy caused by changes in temperature.



Melting

- When a solid is heated, its particles get more energy and vibrate more. This makes the solid expand. At melting point the particles vibrate so much that they break away from their position and solid turn to liquid.

Evaporation or boiling

- When a liquid is heated, its particles get more energy and move faster. They bump into each other more often, and bounce further apart. This makes the liquid expand. At the boiling point, the particles get enough energy to overcome the forces between them. They break away to form a gas.
- Some particles in a liquid have more energy than others. Even well below the boiling point, some have enough energy to escape and form a gas. This is called evaporation.

Condensation

- As a gas cools, its particles lose energy and move more slowly. The particles have less kinetic energy and move closer together. The gas changes state to form a liquid.

Freezing

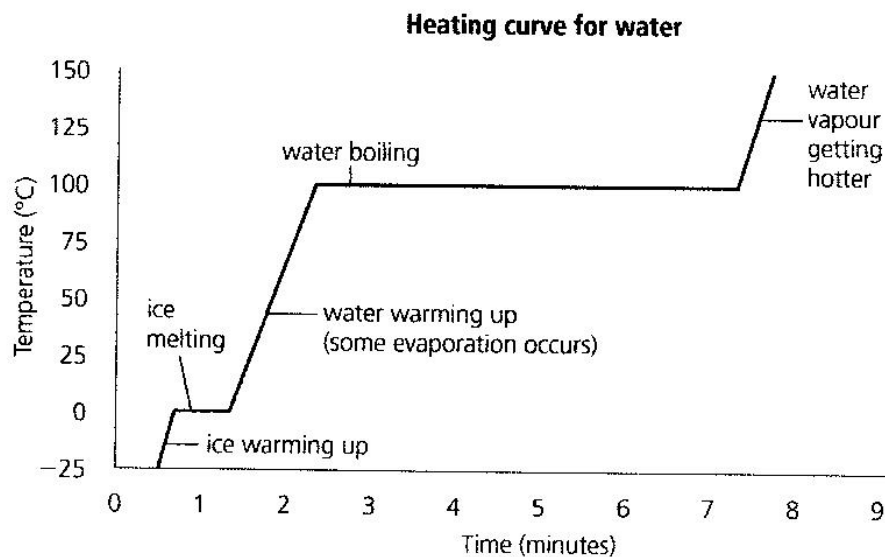
- When a liquid is cooled the particles have less kinetic energy. The particles in a liquid move slower and closer to each other. The liquid changes state to form a solid.

Sublimation and deposition

- With heating, some matter can change from solid state into the gas state, it is called sublimation. When cooled, the gas changes directly to a solid, this is called deposition.

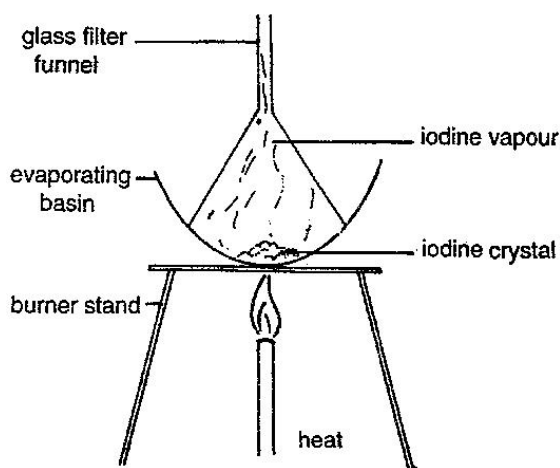
Experiment on heating water

- The temperature remains constant at 0°C as the ice melts. This is because heat supplied is being used to break bonds and change state from solid to liquid (latent heat of fusion).
- When all the ice has melted the temperature rises from zero to 100°C. Particles gain energy and move faster until they break away completely from each other.
- When boiling point is reached, energy being transferred is used to separate the liquid particles from each other.
- As heating continues, all the liquid is changed into a gas. While this is happening the temperature remains constant since heat absorbed is being used to break bonds in a liquid (latent heat of vaporisation)
- The temperature rises again after reaching boiling point. All the particles in the gas are now free to move and further heating raises temperature of the gas.



Experiment on heating iodine

- If iodine is heated in a crucible, it sublimes (sublimation). When the heat source is removed and the crucible is allowed to cool, the iodine cools and become deposited as a solid (deposition).
- Grey iodine changes directly to violet or purple vapor which collects on the cold part of the glass to form black shining crystals of iodine.
- Iodine vapor produced is toxic (poisonous) therefore the experiment should be done in a fume cupboard or close to a window.



SOLUBILITY

- The solubility of a *solute* is the maximum amounts of solute that can dissolve in a certain amount of *solvent* at a certain temperature until it become saturated.

- This means that only so much of a solute will dissolve and any more solute added after saturation will not dissolve even if the particles are further broken down or the solution is heated or stirred.
 - ✓ Solute is any substance that is dissolved in a liquid, to form a *solution*.
 - ✓ Solvent is a liquid that can dissolve other substances to form a solution.
 - ✓ Solution is a homogeneous mixture, which may be liquid, gas or solid, formed by dissolving one or more substances.

Factors that affect solubility

- The rate at which a solute will dissolve depends on particle size, temperature and how much you stir or shake the solution.

Temperature

- An increase in temperature of the solution, increases the solubility of the solute. A solute dissolve more quickly in hot water than it does in cold water because kinetic energy of particles increases as temperature increases thus increasing rate of solubility.

Stirring

- Particles dissolve more quickly when they are stirred because stirring increases the kinetic energy of particles causing them to collide more often with the solvent. Stirring therefore, allows the solute to dissolve faster.

Particle size

- Refers to how big the particles are e.g. fine and coarse salt. Small particles dissolve more quickly than larger ones because a larger surface area of the solute is in contact with the solvent. Many smaller particles will therefore dissolve faster than one large particle which has a smaller surface area in relation to its size. When the total surface area of the solute is increased (breaking a solute into small pieces), the solute dissolves more rapidly.

Elements, mixtures and compounds

Element

- It is a substance made up of one type of atoms and cannot be split into two or more simpler substances by any chemical or physical means. Can exist as either atoms or molecules e.g. Hydrogen, Magnesium, Zinc, Iron, Copper.

Molecule

- It is the smallest electrically neutral particle of an element or compound that can exist on its own e.g. Hydrogen, Nitrogen, and Chlorine, Oxygen, Carbon dioxide and Water.

Mixture

- It is a substance which consists of two or more different substances physical intermingled e.g. salt water or iron filings and sulphur.

Compound

- It is a substance which consists of two or more elements chemically combined together and exists as molecules or ions e.g. copper sulphate, carbon dioxide, magnesium oxide.

Chemical reactions

- Chemical reaction takes place when heat is given out or taken in.
- The burning of magnesium ribbon in air is an example of chemical reaction where heat is taken in and once the reaction starts heat is given out.

Word equations

- Word equations are used to summarize chemical reactions. They show reactants used (left) and products formed (right).

Magnesium + Oxygen (reactants) → Magnesium Oxide (products)

Iron + Sulphur → Iron Sulphide

Physical and chemical changes

- A physical change is one in which the products have the same chemical properties as the reactants (no new substance is formed). Changes are mainly in physical properties or state of matter without accompanying change in composition e.g. evaporation, dissolving.

- A chemical change is one in which the products have chemical properties different from those of the reactants (new substance is formed). Substances are formed with entirely different properties and composition from the original materials e.g. combustion

Differences between physical and chemical changes

Physical Change	Chemical Change
No new substance is formed	New substance is formed
Properties do not change	New substance has different properties
Usually the change is easily reversible	Usually change is irreversible
Usually no energy is given out or absorbed	Usually heat energy is given out or absorbed
No change in mass	There is change in mass
A mixture is formed	A compound is formed

- When wax or ice is heated, no new substances are formed and their masses do not alter. The changes are physical. All changes of states are physical. All changes of states are physical.
- If iron filings and sulphur powder, the iron in the mixture is uncombined with the sulphur and will be attracted to a magnet. But if the mixture is heated, the iron and the sulphur which is not attracted by a magnet. Iron sulphide is a compound which can only be separated by chemical means.
- When Magnesium or mealie meal is heated, a new substance is formed. In many chemical changes, energy is given out as heat, sound or light. It is difficult to change the new substances back to the original Magnesium.

THE PERIODIC TABLE

Classification of elements

- There are 118 elements on the periodic table
- The periodic table is an arrangement of elements in order of their atomic numbers.
- The periodic table shows the names, chemical symbols and atomic numbers for each element.

- The elements are organised into columns (groups) and rows (periods) based on the structure of their atoms and their properties
- The elements in the periodic table are arranged into 3 main categories; metals, metalloids and non-metals.

Groups

- There 8 main groups of elements i.e. I, II, III, IV, V, VI, VII and O. elements between group II and III i.e. group 3-12 are called transition elements.
- Elements in the same group typically have the same number of electrons in their outermost electron shell e.g. elements with one electron in their outermost shell are in group 1.
- Since electrons in the outermost shell are usually the ones that take part in a chemical reaction, elements in the same group undergo the same chemical reactions.
- Some groups have special names;
 - ✓ Group I – alkali metals
 - ✓ Group II – alkaline earth metals
 - ✓ Group VII – halogens
 - ✓ Group VIII – noble gases

Periods

- Periods are numbered 1 to 7
- Electrons in an atom are placed in electronic shells and each shell can contain a separate number of electrons.
- Periods shows the number of shells the elements have. For all elements in the same period, electrons are being added to the same shell as we move across the period from left to right.

Metals and non-metals

- Metals are mostly found on the left and in the middle of the periodic table.
- The elements around the dividing lines are referred to as metalloids and have properties that are similar to both metals and non-metals.
- Non-metals are found on the right of the periodic table

- The zigzag line separates metal from non-metals, with the non-metals on the right of the line, except for Hydrogen. So there is a change from metal to non-metal, as you go across a period.

Common elements

- Each element has its own chemical symbol and below is a list of the first 20 elements of the periodic table based on their atomic number and their symbols.
- Each element has its own name, symbol, atomic number and its own position on the periodic table

Group	I	II	III	IV	V	VI	VII	VIII
Period								
1	${}_1\text{H}$							${}_2\text{He}$
2	${}_3\text{Li}$	${}_4\text{Be}$	${}_5\text{B}$	${}_6\text{C}$	${}_7\text{N}$	${}_8\text{O}$	${}_9\text{F}$	${}_{10}\text{Ne}$
3	${}_{11}\text{Na}$	${}_{12}\text{Mg}$	${}_{13}\text{Al}$	${}_{14}\text{Si}$	${}_{15}\text{P}$	${}_{16}\text{S}$	${}_{17}\text{Cl}$	${}_{18}\text{Ar}$
4	${}_{19}\text{K}$	${}_{20}\text{Ca}$						

CONCENTRATIONS

- Concentration is the amount of solute that is dissolved in a solvent.
- If the identity of the solute and solvents in a solution are known, the concentration of the solution can be determined.

Experiment to determine concentration by colour intensities

- Put KMnO_4 crystals in test tube A and add 20cm^3 of water. Stir the solution until all crystals have dissolved.
- Use a pipette to measure 2cm^3 of the solution in test tube A and transfer it to test tube B. Add 18cm^3 of water.
- Repeat for test tube C and D, each time taking 2cm^3 of solution from the previous test tube.
- The colour intensities of the various solutions will range from dark or deep purple colour to a light or very light pink colour because a 1:10 dilution was created each time since

each successive concentration is $\frac{1}{10}$ of the previous concentration. The final dilution appears much lighter in colour than the others.

- To determine the % concentration of a solution, the following equation is used;

$$\% \text{ composition} = \frac{\text{total volume of solution}}{\text{volume of solvent}} \times 100$$

- Higher concentration of a substance will give solutions of darker or deeper colours. As the solutions are diluted, their intensities decrease.

ACIDS, BASES AND SALTS

ACIDS

- An acid is a compound which when dissolved in water, forms hydrogen (H^+) ions as the only positively charged ions.
- Strong acids are completely ionized e.g. sulphuric acid, Hydrochloric acid and Nitric acid. Weak acids are only slightly ionized in dilute solutions.

- Examples of acids hydrochloric acids (HCl), sulphuric acids (H₂SO₄) and nitric acid (HNO₃)

Properties of Acids

- Turn litmus paper red, methyl orange red, phenolphthalein colorless and universal indicator red or orange
- Have a pH below 7
- They are corrosive
- Have a sour taste

BASES

- A base is a compound which contains oxide or hydroxide ions.
- An alkali is a compound which when dissolved in water forms hydroxide (OH⁻) ions. They dissolve in water to form alkaline solutions,
- Examples of bases all metal oxides, all metal hydroxides and ammonia

Properties of Bases

- Turn litmus paper blue, universal indicator blue or purple, phenolphthalein pink and methyl orange yellow.
- Taste bitter
- Have a slippery feel
- Strong bases are caustic e.g. potassium hydroxide

Identifying acids and bases using litmus paper

- Litmus is a purple dye that can be in form of a solution or in the form of red or blue paper
- Litmus is called an indicator because it indicates whether a substance is a base or an acid.
- Substances that are neutral are neither acidic nor basic.
- Both red and blue litmus paper should be used to test a solution. If the colour does not change, it means that it does not indicate that the substance is an acid or base
- Dip litmus paper in five test tubes with each of the following substances i.e. sodium hydroxide, dilute hydrochloric acid and distilled water, tap water, copper sulphate
- Record the results in the table below and draw conclusions about the colour change in each substance.

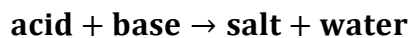
Substance	Effect on red litmus paper	Effect on blue litmus paper	Acid or alkali
Hydrochloric acid (HCl)			
Sodium hydroxide (NaOH)			
Tap water			
Distilled water			
Copper sulphate solution			

Acid – Base reactions

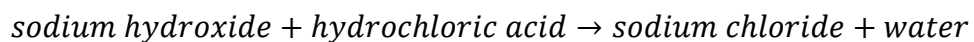
- When an acid react with a base, it is a neutralization reaction.
- During the reaction, salt is always produced. Type of salts produced depends on the acid used i.e.

Acid	Salt formed
Hydrochloric acid (HCl)	Chloride
Sulphuric acid (H ₂ SO ₄)	Sulphate
Nitric acid (HNO ₃)	Nitrate

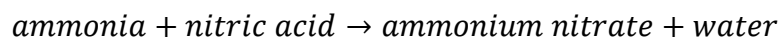
- Bases release hydroxide ions (OH⁻) while acids release (H⁺). During neutralization H⁺ ions combine with OH⁻ ions to form water



- A reaction between sodium hydroxide and hydrochloric acid gives sodium chloride and water



- A reaction between ammonia and nitric acid give ammonium nitrate and water



- A reaction between copper oxide and sulphuric acid gives copper sulphate and water

copper oxide + sulphuric acid → copper sulphate + water

INDUSTRIAL PROCESSES

PRODUCTION OF PEANUT BUTTER

Shelling

- Consists of removing the peanut shell with the least damage to the seeds inside.
- The peanuts are passed between a series of rollers where the peanut shells are gently cracked. The cracked peanuts are then repeatedly passed over screens and blowers, where they are shaken, gently tumbled and air-blown until all the shells are removed.
- Another option would be to shell the peanuts by hand and using a winnowing basket to separate the lighter shells from the peanuts

Roasting

- Peanuts are roasted in special ovens at 180°C for about 10 minutes. This process destroys certain enzymes in the peanuts that may produce bad flavors. Roasting also enhances the color, flavor and texture of the peanuts.
- Roasting at home is done using a pan over a wood fire or stove.

Grinding

- The peanuts are lightly rubbed between rubber belts to remove their outer skins around each peanut. The result is peanuts that are paler in colour than they were with skins. This is called blanching.
- The splitted peanuts are then ground in a grinding machine twice. First time reduces the nuts to a medium grind and the second time produce a fine smooth texture.
- The peanuts are ground alone first and then ingredients are added to them like salt, sweetener and stabilizer. The stabilizer stops the oil from separating out of the butter

- In rural areas splitted peanuts are pounded in a mortar (duri) using a pestle (mutswi) followed by grinding on the grinding stone (guyo).

Packaging

- The peanut butter is cooled and taken to filling machine where the correct amount of peanut butter is poured into empty jars.
- An automatic capping machine places a lid on each jar. The jars are labeled to show the contents mass and nutritional values. The jars are sealed to prevent it from being exposed to the air. They are packed ready to be transported to stores for sale.

Production of oil from peanut butter

- Pour the peanut butter in a bowl and cover the lid tightly and put in a refrigerator for at least a day. The oil will rise on top of the butter.
- Strain the oil into a separate bowl. When the oil is completely clean after several times of straining, pour it in a clean bottle.

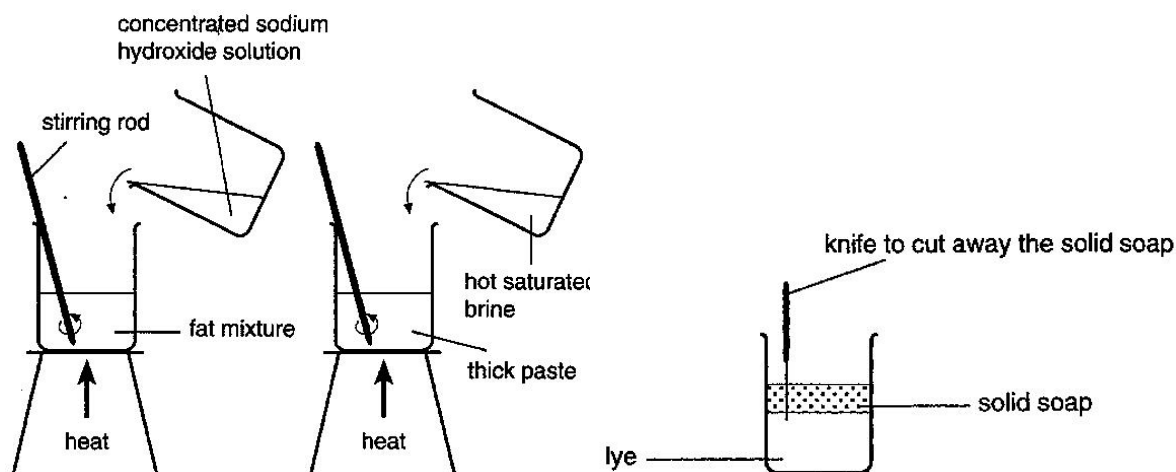
Uses of oil

- Cooking oil – it has a higher burning and smoking point
- Baby care products such as nappy rash cream
- Baking
- In salad making
- Skin moisturizer
- Massage oil – it has a light and nutty smell
- Healthy oil because it can be produced at low temperature.

PRODUCTION OF SOAP

Saponification

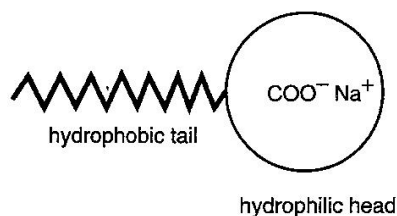
- Soaps are made by the action of alkalis on fatty acids, a process called *saponification*.



- Oils and fats are heated by steam with sodium hydroxide. The fats are broken down, leaving glycerol and the sodium salt of the long chained acid.
- The sodium salt of the acid can be used as soap.
- Sodium chloride is added to separate the soap as an upper layer (the salting out of soap).



- The soap molecule has a hydrophilic head and the hydrocarbon tail is hydrophobic. When washing with soap and water, the hydrophilic end is attracted to the water molecules while the hydrophobic end mix with the oils from the dirty object, detaching them from the object and attach themselves to the soap



Stages in the production of soap

Boiling

- The alkali and fat are boiled in a large kettle (steel tank). After boiling the mass thickens as the fat reacts with the alkali producing soap and glycerine

Separation or Salting

- Salt is added to the mixture to cause the soap product to rise to the top of the kettle and allow the glycerol to settle at the bottom. In this way the glycerol is simply tapped off from the bottom of the kettle

Refinement

- To remove the small amounts of fat that have not saponified, a strong caustic alkali solution is added to the kettle. The mass is brought to the boil again and react the last of the fat turns to soap

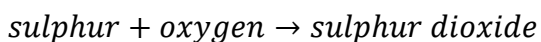
Pitching

- Involves boiling the soap again with added water. The mass eventually separates into two layers. The bottom layer called *nigre* containing most of the water, impurities that such as dirt and salt. The top layer, called the *neat soap* is a liquid layer that contains all the purified soap (70%) and some water (30%). The neat soap is poured into moulds and allowed to cool.

OXIDATION AND REDUCTION

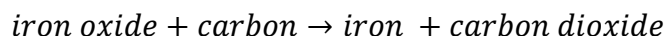
Oxidation

- Oxidation is gain of oxygen
 - Sulphur is oxidized to sulphur dioxide by gaining oxygen.



Reduction

- Reduction is the removal of oxygen
 - Iron oxide is reduced to iron by losing oxygen to carbon



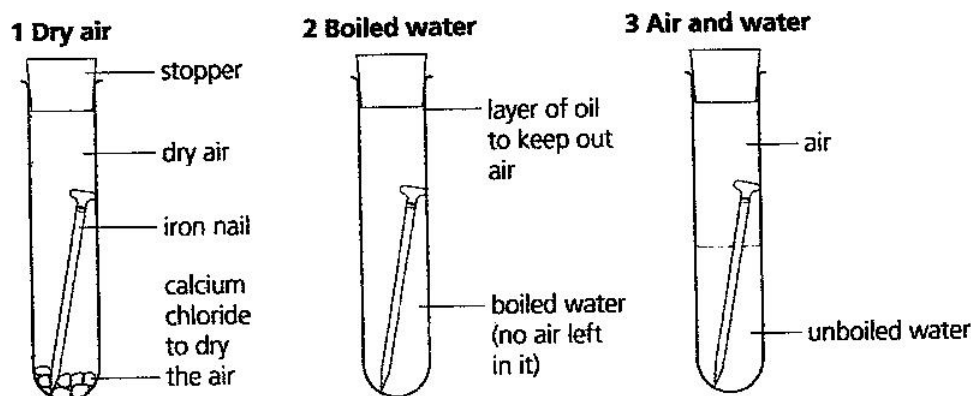
Rusting

- Rusting is the corrosion of iron i.e. the formation of iron oxide.
- When iron comes into contact with oxygen, a chemical reaction takes place, producing iron oxide (rust)
- When the rust forms, the flakes of rust fall off the surface of the metal, exposing more iron to the factors that cause rusting, resulting in further rusting of the object. In this way, the iron is gradually corroded hence deteriorates and destroyed over time by rust.

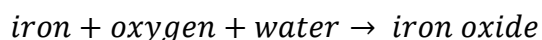
Conditions necessary for rusting

- Water
- Oxygen

Experiment on conditions needed for rusting



- Stand three identical nails in three test tubes.
- Prepare the test-tubes as below so that
 - Test tube 1 - Contains dry air
 - Test tube 2 - Contains water but no air
 - Test tube 3 - Has both air and water
- Leave the test-tubes for several days
- After several days the nails in test 1 and 2 shows no signs of rusting. The nail in test tube 3 has rust on it. This is because rusting requires water and oxygen. In fact the iron is oxidized in this reaction.
- The anhydrous calcium chloride in test tube 1 absorbed moisture thus drying the air.
- The water in test tube 2 was boiled to remove air from it. A layer of oil prevents the entry of air into the tube.



Methods of preventing rusting

1. Surface protection – covers the metal with a layer of substance e.g.
 - Painting or oiling – involves applying a thin coat of paint which will not allow both air and water from coming into contact with the metal. The paint must be kept in good condition because any scratch in the paint surface will expose the iron to air and moisture causing it to rust.
2. Sacrificial protection – covering metal with thin layer of another more reactive metal e.g.
 - Electroplating – iron is often coated with other metals to prevent it from rusting e.g. plating with copper, chromium and nickel.

- Galvanizing – iron or steel is coated with zinc by dipping the sheet of iron in molten zinc. Zinc corrodes before iron but even when corroded it forms a layer of zinc oxide which prevents the iron from being corroded.

ORGANIC CHEMISTRY

FUELS

- A fuel is a material that can be burnt to give out heat energy or have chemical energy.

Types of fuels

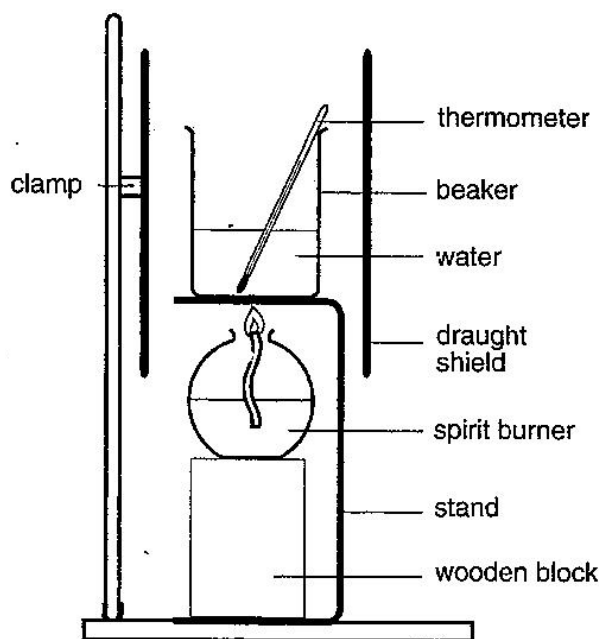
- Fuels are found in three forms i.e.
 - **Solid fuels** e.g. wood, charcoal, coke and coal. They are relatively cheap and mostly readily available. They are easy to store. Have low heating efficiency. Extraction can cause environmental degradation and have high carbon emissions.
 - **Liquid fuels** e.g. petrol, diesel, paraffin and ethanol. More expensive than solids. Are less available and can be stored more compactly and in a closed container. High heating efficiency and are more flammable than solid fuel. Release carbon emissions when burnt.

- **Gaseous fuels** e.g. methane, ethane, hydrogen and coal gas. Costly and less available. Needs to be stored in leak proof container. High heating efficiency and have very risk, incredibly flammable and leaks can go undetected. Burns without smoke but fracking for natural gas causes land and water pollution.

Thermal efficiency

- Thermal efficiency is the measure of the heat content of a fuel.
- Carbon in fuels burns to produce heat energy.
- The efficiency of a fuel can be measured by the rate at which the fuel heats up a substance to certain temperature.

Comparing the efficiency of different fuels



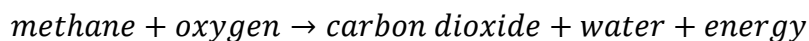
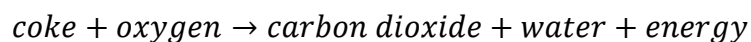
- Equal volumes of liquid fuels were used to heat equal volumes of water for the same period of time.
- Temperature readings were taken at the beginning and at the end of the heating process.
- The temperature of water heated using methylated spirits increased less than that of paraffin.
- Paraffin has a higher thermal efficiency than methylated spirit. This is because paraffin has a higher carbon content which is evident because the flame has more soot.

- The more the carbon content the fuel contains, the higher the heating efficiency e.g. coke and charcoal have higher carbon content than wood and coal and therefore give out more heat energy when burnt.

Type of fuel	Examples	Heating value
Solid	Wood	17
	Coal	25
	Coke	28
	Charcoal	33
Liquid	Ethanol	30
	Petrol	45
	Paraffin	48
	Diesel	55
Gas	Coal gas	43
	Biogas	40
	Butane	50
	Methane	55
	hydrogen	60

Complete and Incomplete combustion

- Combustion means the burning of a fuel to give energy.
- Many fuels contain carbon and hydrogen. When the fuels burn the carbon is oxidized to carbon dioxide while hydrogen is oxidized to water.
 - Complete combustion occurs when a fuel burns in a plentiful supply of oxygen. All carbon is burnt.

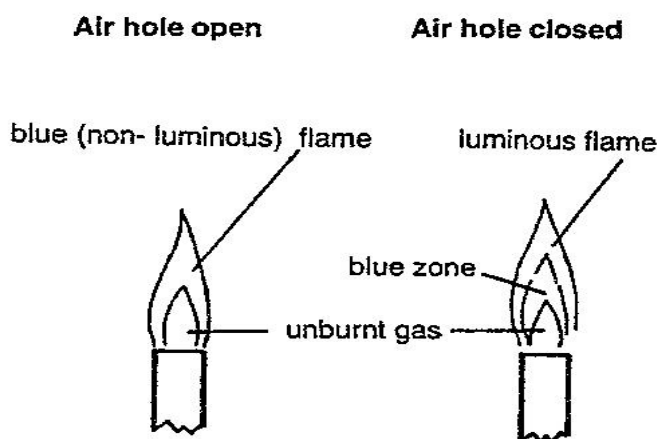


- Incomplete combustion occurs when the amount of carbon in a fuel exceeds the amount of oxygen available. Not all carbon is burnt is given off as soot while some has partial oxygen and is turned into carbon monoxide. Carbon monoxide is

a poisonous gas which can kill if excessively inhaled. Incomplete combustion causes a lot of air pollution because of the soot and carbon monoxide.

paraffin + little oxygen $\rightarrow$ carbon monoxide + soot + water + little heat

Demonstrating complete and incomplete combustion using burners



- The blue zone on the Bunsen burner indicates complete combustion and the yellow flame indicates incomplete combustion.
- The air hole on the burner controls the amount of oxygen that mixes with the gas.
- When air hole is open, air enters the tube and mixes with the gas which therefore burns quickly and completely. The flame is small and has a dominant non-luminous blue flame. The flame is clean and very hot and can heat water very rapidly.
- When the air hole is closed, no air enters the tube and the gas mixes with little air hence incomplete combustion. The flame is larger and has a dominant yellow luminous flame which is sooty which blackens the underside of the beaker. The flame is not very hot and gives little energy such that it takes longer to raise temperature of water

Effects of complete and Incomplete Combustion

1. Deforestation

- Uncontrolled cutting down of trees for firewood causes deforestation which will result in soil erosion and loss of top soil causing river siltation.

- Deforestation upsets the balance of nature i.e. plant and animal habitats are destroyed so that they cannot support the normal animal population.
- Deforestation causes an increase in the carbon dioxide level in the atmosphere because there is now less vegetation to remove atmospheric carbon dioxide during photosynthesis.

2. Pollution

- Incomplete combustion of fuels like petrol in motor engines leads to emission of soot and carbon monoxide. Carbon monoxide is a poisonous gas while soot causes smog over cities.
- Fuels also give off volatile substances which pollute the air most of which may cause cancer.
- The nitrogen and sulphur from fossil fuels are oxidised into acidic oxides that make rain water more acidic than it should be. Plants are damaged by acid rain, fish in lakes and rivers cannot tolerate acidic conditions hence are destroyed.

3. Global Warming

- Large amounts of carbon dioxide build up and a layer of gas is created in the atmosphere which stops heat from escaping the atmosphere and is reflected back to earth resulting in global warming. Rise in earth's temperature will lead to fertile land becoming dry and uninhabitable. In arctic regions ice caps slowly melts resulting in flooding.